Chapter 7 exercises

DS07-01 • Stability and inversion count

Run stable insertion sort on `(5,A),(2,B),(4,C),(2,D)`. List each pass, the initial strict inversion count, and shift count. What fails if the movement condition becomes greater-than-or-equal?

DS07-02 • Three-way partition and stack depth

Trace lt, i, gt on `[4,2,4,7,1,4]` with pivot 4 using the stated partition algorithm. Prove that recursing on the smaller side and looping on the larger limits stack depth to O(log n), but does not ensure O(n log n) worst-case time.

DS07-03 • The model behind the lower bound

Prove the worst-case Ω(n log n) comparison-sorting bound for distinct keys using a decision tree. Explain why counting sort does not contradict it and why the n! argument cannot directly describe an all-equal-key input.

DS07-04 • A counterexample to unstable LSD passes

Apply decimal LSD radix sort to `31,14,22,11`. Suppose the second pass reverses lower-digit order among equal tens digits. Give the incorrect output. Compare the costs of a very large base b and a very large digit count d.

References and attribution

Ellis Horowitz, Sartaj Sahni, Dinesh Mehta. Fundamentals of Data Structures in C++, 2nd Edition.

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https://github.com/xinhi6716/computer-science-self-study